Ex 9.4 Q47

$-\frac{\pi}{2} < \cancel{\tan}\ \arctan x < \frac{\pi}{2}$

$(x \in \mathbb{R})$

$$\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^{1.1}} \to n = 1, 2, 3, \ldots \text{ (positive)}$$

If $n \in \mathbb{N}$, $0 < \arctan n < \frac{\pi}{2}$.
$(n = 1, 2, 3, \ldots)$

$$\Rightarrow \boxed{0 <} \frac{\tan^{-1} n}{n^{1.1}} < \frac{\pi}{2} \cdot \underbrace{\frac{1}{n^{1.1}}}$$

Since $1.1 > 1$, $\sum \frac{1}{n^{1.1}}$ conv.

$\Rightarrow \sum \frac{\pi}{2} \cdot \frac{1}{n^{1.1}}$ conv.

$\frac{\pi}{2} \sum \frac{1}{n^{1.1}}$ conv.

$(1.1 > 1)$

$\Rightarrow \sum \frac{\tan^{-1} n}{n^{1.1}}$ conv. by DCT.

$$\coth n = \frac{1}{\tanh n} \to 1$$

$$\sinh n := \frac{e^{n} - e^{-n}}{2}$$

$$\cosh n := \frac{e^{n} + e^{-n}}{2}$$

$$\tanh n = \frac{\sinh n}{\cosh n}$$

$$= \frac{e^{n} - e^{-n}}{e^{n} + e^{-n}} \cdot \frac{\frac{1}{e^{n}}}{\frac{1}{e^{n}}}$$

$$= \frac{1 - e^{-2n}}{1 + e^{-2n}} \to 1$$

$$\sqrt[n]{n} \to 1$$

$$n^{1/n}$$

$$\operatorname{sech} n = \frac{1}{\cosh n} = \frac{2}{e^{n} + e^{-n}} \to 0$$

Q49 $\displaystyle\sum_{n=1}^{\infty} \frac{\boxed{\coth n}}{n^2}$ → This seq. conv. to 1

$\approx \frac{1}{n^2}$ (LCT)

Let $a_n = \dfrac{\coth n}{n^2}$, $b_n = \dfrac{1}{n^2}$ for $n \geq 1$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \coth n = \underbrace{\boxed{\cdots}}_{\text{Done just now}} = 1 \underline{> 0}$$

Since $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ conv., $\displaystyle\sum_{n=1}^{\infty} \frac{\coth n}{n^2}$ conv.

by LCT.

Extra

$$\sum \left(\frac{n+1}{n\sqrt[n]{\ln n}}\right)^n$$

$$\underbrace{\left(\frac{n+1}{n}\right)^n}_{\left(1+\frac{1}{n}\right)^n \to e} \cdot \frac{1}{\ln n} \approx \boxed{\frac{1}{\ln n}}$$

$\ln n \leq n$

$\frac{1}{n} \leq \frac{1}{\ln n}$

Integral Test / Direct Comparison Test

$\sum \frac{1}{\ln n}$ diverges

(LCT) $a_n = \left(\frac{n+1}{n}\right)^n \cdot \frac{1}{\ln n}$. $b_n = \frac{1}{\ln n}$

$$\lim \frac{a_n}{b_n} = \cdots = e > 0$$

Ex 9.5 Q23

constant

① $$\sum \frac{2+(-1)^n}{(1.25)^n} = \sum \frac{\boxed{2}}{(1.25)^n} + \sum \frac{(-1)^n}{(1.25)^n}$$

$$= 2\sum \left(\frac{1}{1.25}\right)^n + \sum \left(-\frac{1}{1.25}\right)^n$$

$$\left|\pm\frac{1}{1.25}\right| < 1 \Rightarrow \sum \left(\frac{1}{1.25}\right)^n \text{ and } \sum\left(-\frac{1}{1.25}\right)^n$$

are conv.

Write before $\sum(\quad) = \underline{\sum(\ )} + \underline{\sum(\ )}$

$\therefore \sum \sim$ conv.

② (Root)

$$\sum \frac{2+(-1)^n}{(1.25)^n}$$

$$a_n = \frac{2+(-1)^n}{(1.25)^n} \text{ for } n \geq 1$$

$\sum a_n$ conv. by Root Test

$\Uparrow$

$$\sqrt[n]{|a_n|} = \frac{\sqrt[n]{2+(-1)^n}}{1.25} \to \frac{1}{1.25} < 1$$

$$1 \leq 2+(-1)^n \leq \underline{3} \Rightarrow \underline{1} \leq \sqrt[n]{2+(-1)^n} \leq \underbrace{\sqrt[n]{3}}_{\to 1}$$

$$\Rightarrow \sqrt[n]{2+(-1)^n} \to 1 \text{ by Sqz. Thm.}$$

$\lim |a_n| \neq 0 \Rightarrow \lim a_n \neq 0$

$\Leftrightarrow \underline{\lim a_n = 0} \Rightarrow \lim |a_n| = 0$

(Contrapositive)

$P \Rightarrow Q \equiv \sim Q \Rightarrow \sim P$

$\downarrow$

Prove using ε-N Language

Let $\varepsilon > 0$.

Idea: Find $N > 0$ s.t. $\underbrace{||a_n||}_{=|a_n|} < \varepsilon$ for all $n \geq N$.

Ex. 9.6 Q 20

$$\boxed{\lim \frac{x^n}{n!} = 0 \quad \forall x \in \mathbb{R}}$$

$$\sum (-1)^{n+1} \frac{n!}{2^n}$$

$$a_n := (-1)^{n+1} \frac{n!}{2^n}$$

$$|a_n| = \frac{n!}{2^n} \to \infty$$

$$\frac{n!}{2^n} = 1 \Big/ \left(\frac{2^n}{n!} \right)$$

$\because |a_n| > 0$

$$\frac{2^n}{n!} \to 0$$

$\therefore \underline{\lim |a_n| \neq 0}$

$\Rightarrow \lim a_n \neq 0$

$\sum |a_n|$ div.

By n-term test,

$\sum (-1)^{n+1} \frac{n!}{2^n}$ div.